Yonathan T. Gashu

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Education

Georgia Institute of Technology Atlanta, GA

Bachelor of Science in Computer Science - GPA 3.90 (Major) / 3.88 (Overall) *Graduating May 2027*

- Specializing in *Systems & Architecture* and *Devices*
- Coursework: Design of Operating Systems, Compilers, Computer Architecture, Computer Networking, Algorithms

Experience

Department of Defense Aug. 2023 - Present

Capabilities Development Specialist (Computer Network Exploitation Development), Oahu, HI May 2026 - Aug. 2026

- Conducted vulnerability analysis across multiple versions of target software and validated AI-generated proof of concept exploits through testing and debugging
- Developed multi-stage attack paths demonstrating pre-authenticated and authenticated remote code execution capabilities
- Identified and documented risks (artifacts, logging signatures, unencrypted payloads) to inform security considerations

Capabilities Development Specialist (Control Networks Section), Fort Meade, MD May 2025 - Aug. 2025

- Employed reverse engineering techniques to examine network-enabled embedded devices and exploit firmware vulnerabilities
- Developed proof-of-concept tools enabling remote code execution
- Conducted static and dynamic analysis on control flow of functions of interest to enhance understanding of device runtime behavior
- Authored comprehensive documentation outlining findings and usage of created capabilities

Computer Systems Researcher (Artificial Intelligence Research Office), Fort Meade, MD May 2023 - Aug. 2024

- Worked alongside senior researchers to investigate the extraction of knowledge graphs (KGs) from unstructured text
- Collaborated on the development of a KG-RAG system to enable Knowledge Graph Question Answering

Software Engineer Intern (Data Targeting Solutions Division), Fort Meade, MD Sep. 2022 - Aug. 2023

- Utilized Java to develop backend database management features for an internal web application and successfully integrated those functionalities into the frontend interface using VueJS
- Created a developer-sided API to streamline the creation of test agreements/datafeeds

Cyber-Physical Security Lab (CPSec) Jan. 2026 - Present

Undergraduate Researcher, Binary Analysis for Bill of Materials (BABOM) *Atlanta, GA*

- Developing ground-truth methodology for identifying and slicing third-party library code out of statically linked firmware binaries using function call graph extraction
- Achieved ~99% function recovery on unstripped and ~93% on stripped curl/OpenSSL binaries (*angr, pyhidra, Ghidra*)

Projects

Requiem | *Rust, x86-64, Intel VT-x* Mar. 2026 - Present

- Building a Type-1 bare-metal hypervisor in Rust for x86-64 using Intel VT-x, booting from UEFI
- Implemented VMX enablement: page-aligned VMXON region from a bump allocator over UEFI-allocated memory and a verified `vmxon` entry into VMX root operation; VMCS setup in progress

xv6 Kernel Extensions (CS3210) | *C, x86* Aug. 2026 - Present

- Implemented copy-on-write fork and lazy zero-page allocation in the xv6 virtual memory subsystem
- Added kernel backtrace, a `setscheduler` syscall, and `clone`-based threading

IP Camera Vulnerability Research | *Embedded, Firmware RE* Apr. 2026 - Present

- Gained a root shell on a Uniview IP camera through UART and U-Boot, and extracted the full filesystem for offline analysis
- Mapping the device's network attack surface toward a remotely exploitable vulnerability and CVE disclosure

LoRa Enabled Wearable Off-grid Communication (LEWOC) | *Rust, Embedded Systems* Aug. 2025 - Dec. 2025

- Built an encrypted LoRa text-messaging wearable in `no_std` Rust/Embassy on RP2350 with Ascon encryption, a TFT display UI, and Bluetooth app connectivity, achieving 1000-foot range

Technical Skills

Languages: C, C++, Rust, Python, Java, C#, Assembly (ARM64, x86-64)
Reverse Engineering: Ghidra, Binary Ninja, GDB, angr, pyhidra, Wireshark
Systems & Tools: Linux, QEMU, KVM, UART/U-Boot, Git, Docker, VMware